

Generic ranks of odd-order Hankel tensors from three compressed slices

Recovered candidate manuscript for TR-13. Not independently refereed.

Abstract

For every odd order $m \geq 3$ and dimension $n \geq 2$, the generic ordinary rank, symmetric rank, ordinary border rank, symmetric border rank and Vandermonde rank of a complex Hankel tensor all equal $\lceil (m(n-1) + 1)/2 \rceil$. The lower bound is certified by an explicit three-slice Koszul flattening after grouping the tensor factors. Its rank is established using a two-spike generator and a nilpotent-shift commutator. This addresses the generic odd-order statement in TR-13; no claim is made about equality of ranks for every Hankel tensor.

1 Statement and notation

Use zero-based indices. For $h = (h_0, \dots, h_D) \in \mathbb{C}^{D+1}$, where $D = m(n-1)$, write

$$H(h)_{i_1, \dots, i_m} = h_{i_1 + \dots + i_m}, \quad 0 \leq i_j < n.$$

Let R , R_{sym} , $\underline{R}$, $\underline{R}_{\text{sym}}$ and R_V denote ordinary rank, symmetric rank, ordinary border rank, symmetric border rank and Vandermonde rank. A Vandermonde summand is a scalar multiple of $(1, t, \dots, t^{n-1})^{\otimes m}$, or its projective counterpart at infinity. The ordinary border rank is taken in the full tensor space. The conventions and the earlier even-order and cubic results are in Nie–Ye [1]; the remaining generic odd-order assertion is the canonical TR-13 target [2].

Theorem 1. *Fix an odd integer $m \geq 3$ and an integer $n \geq 2$. There is a nonempty Zariski-open subset of the Hankel tensor space on which*

$$R(H) = R_{\text{sym}}(H) = \underline{R}(H) = \underline{R}_{\text{sym}}(H) = R_V(H) = \left\lceil \frac{m(n-1) + 1}{2} \right\rceil.$$

Put

$$m = 2k + 1, \quad \ell = n - 1, \quad a = k\ell + 1, \quad s = \lfloor \ell/2 \rfloor, \quad r = a + s.$$

One checks separately for even and odd ℓ that $r = \lceil (D+1)/2 \rceil$. The proof consists of a generic Vandermonde upper bound and a matching ordinary-border-rank lower bound.

2 An elementary three-slice rank certificate

Lemma 2. *Let $T \in \mathbb{C}^a \otimes \mathbb{C}^3 \otimes \mathbb{C}^a$ have slices M_0, M_1, M_2 in its middle factor. Define the $3a \times 3a$ matrix*

$$K(T) = \begin{pmatrix} -M_1 & M_0 & 0 \\ -M_2 & 0 & M_0 \\ 0 & -M_2 & M_1 \end{pmatrix}.$$

Then $\underline{R}(T) \geq \lceil \text{rank } K(T)/2 \rceil$. If M_0 is invertible, then

$$\text{rank } K(T) = 2a + \text{rank}(M_1 M_0^{-1} M_2 - M_2 M_0^{-1} M_1).$$

Proof. For a rank-one tensor, all slices have the form $M_j = c_j uv^\top$. The displayed block matrix is the tensor product, up to ordering, of the rank-one matrix $uv^\top$ and

$$\begin{pmatrix} -c_1 & c_0 & 0 \\ -c_2 & 0 & c_0 \\ 0 & -c_2 & c_1 \end{pmatrix},$$

whose rank is at most two. By subadditivity, $\text{rank } K(T) \leq 2R(T)$. Every $(2q+1)$ -minor of K is a polynomial in the tensor entries and vanishes on all tensors of rank at most q . It therefore vanishes on their closure. This proves the border-rank bound without any symmetry restriction on an approximating sequence.

When M_0 is invertible, (x, y, z) belongs to $\ker K$ exactly when

$$\begin{aligned} y &= M_0^{-1} M_1 x, & z &= M_0^{-1} M_2 x, \\ (M_1 M_0^{-1} M_2 - M_2 M_0^{-1} M_1) x &= 0. \end{aligned}$$

The formula follows by computing the nullity. $\square$

3 Compression of the Hankel tensor

Group the first k factors, the middle factor, and the last k factors. For every $u \in \{0, \dots, k\ell\}$ choose once and for all a multi-index $I(u) \in \{0, \dots, \ell\}^k$ with sum u . Such an index exists by successively allocating at most ℓ to each coordinate. Selecting the coordinates indexed by $I(u)$ is a linear map from the grouped factor $\mathbb{C}^{n^k}$ onto $\mathbb{C}^a$.

Applying this map to both grouped factors gives a tensor of format $a \times n \times a$ with entries h_{u+j+v} . For $n \geq 3$, select the middle coordinates $0, s, 2s$, obtaining $T(h) \in \mathbb{C}^a \otimes \mathbb{C}^3 \otimes \mathbb{C}^a$ with slices

$$(M_j)_{uv} = h_{u+v+js}, \quad 0 \leq u, v < a, \quad j = 0, 1, 2. \quad (1)$$

Grouping and applying fixed linear maps send a rank-one original tensor to a rank-one tensor or zero. They are also continuous. Hence

$$\underline{R}(H(h)) \geq \underline{R}(T(h)). \quad (2)$$

This step does not require a structured decomposition of $H(h)$.

4 A two-spike generator

Assume $n \geq 3$, so $s \geq 1$. Choose

$$h_{a-1} = h_{2a+s-1} = 1, \quad h_i = 0 \text{ for all other } i. \quad (3)$$

Both indices lie in $\{0, \dots, D\}$: the only nontrivial check is $2a + s - 1 \leq D = 2a - 2 + \ell$, which is equivalent to $s + 1 \leq \ell$. The indices are distinct. Moreover $a \geq 2s + 1$, since $k \geq 1$ and $2s \leq \ell$.

Let J be the reversal matrix, $J_{uv} = 1$ when $u + v = a - 1$ and zero otherwise, and let L be the lower shift $Le_i = e_{i+1}$ for $i < a - 1$, $Le_{a-1} = 0$. Define

$$E = \sum_{i=0}^{s-1} e_i e_{a-s+i}^\top.$$

Substitution of (3) into (1) gives

$$M_0 = J, \quad B := JM_1 = L^s, \quad C := JM_2 = L^{2s} + E. \quad (4)$$

For completeness, the first spike in JM_j occurs at column minus row $-js$, giving L^{js} . The second spike cannot occur for $j = 0$ or $j = 1$; for $j = 2$ it occurs at column minus row $a - s$, giving E .

Since powers of L commute, (4) yields

$$\begin{aligned} CB - BC &= EL^s - L^s E \\ &= \sum_{i=0}^{s-1} \left(e_i e_{a-2s+i}^\top - e_{s+i} e_{a-s+i}^\top \right). \end{aligned} \quad (5)$$

In the first $2s$ rows and last $2s$ columns the matrix in (5) is $\text{diag}(I_s, -I_s)$; all other entries vanish. In particular its rank is exactly $2s$. As $M_0^{-1} = J$, Lemma 2 gives

$$\text{rank } K(T(h)) = 2a + 2s = 2r.$$

Some $2r$ -minor of $K(T(h))$ is thus a nonzero polynomial in h . It is nonzero on a nonempty Zariski-open set. On that set, (2) and Lemma 2 imply

$$\underline{R}(H(h)) \geq r. \quad (6)$$

The special generator is used only to certify a nonzero minor. It is not being assumed to lie in the eventual open set on which all five ranks agree.

If $n = 2$, then $s = 0$ and $r = a = k + 1$. Grouping k factors against $k + 1$ factors and selecting one representative for each index sum produces an $a \times (a + 1)$ ordinary matrix flattening. For the generator $h_k = 1$ and all other entries zero, its first a columns form J . Thus its generic rank is a , proving (6) in the binary case as well.

5 Generic upper bound and conclusion

Consider the polynomial moment map

$$\mathcal{M} : \mathbb{C}^{2r} \longrightarrow \mathbb{C}^{D+1}, \quad ((\lambda_j, t_j)_{j=1}^r) \longmapsto \left(\sum_{j=1}^r \lambda_j t_j^i \right)_{i=0}^D.$$

At distinct nodes with every $\lambda_j \neq 0$, its Jacobian columns are $v_D(t_j)$ and $\lambda_j v'_D(t_j)$, where $v_D(t) = (1, t, \dots, t^D)^\top$. A linear functional annihilating all these columns corresponds to a polynomial f of degree at most D satisfying $f(t_j) = f'(t_j) = 0$ for every j . As $2r \geq D + 1$, such a polynomial is zero. The Jacobian therefore has full row rank $D + 1$.

It follows that $\mathcal{M}$ is dominant. Its image is constructible and contains a nonempty Zariski-open subset of $\mathbb{C}^{D+1}$; equivalently one may apply the standard dominance and constructibility facts for polynomial maps. For every h in this image,

$$H(h) = \sum_{j=1}^r \lambda_j (1, t_j, \dots, t_j^\ell)^{\otimes m},$$

so $R_V(H(h)) \leq r$. Intersect this nonempty open set with the nonempty open set from (6). Their intersection is nonempty because the Hankel parameter space is irreducible. On the intersection, the inequalities

$$\underline{R} \leq R \leq R_{\text{sym}} \leq R_V, \quad \underline{R} \leq \underline{R}_{\text{sym}} \leq R_{\text{sym}}$$

force every displayed rank to equal r . This proves Theorem 1.

6 Scope and reproducibility

Theorem 1 addresses the full generic statement for every admissible pair (m, n) in TR-13. It also supplies a second proof for order three, which was already known [1]. It is not a proof of Comon’s conjecture for all Hankel tensors (TR-14): the nonzero-minor and moment-map arguments both specify open sets and do not eliminate exceptional tensors.

The accompanying exact-arithmetic script `code/verify_tr13.py` checks the slice identities, the entire commutator, and its indicated nonzero minor over a parameter grid. For small formats it independently builds the full Koszul matrix and computes its rank, and verifies that rank-one inputs have Koszul rank at most two. No floating-point rank tolerance enters these checks. The parameter-uniform proof is the one above; the grid is only a diagnostic.

References

- [1] J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM Journal on Matrix Analysis and Applications 40 (2019), doi:10.1137/18M1168285. Author preprint, Sections 4–6, especially Question 6.1 and Conjecture 6.2: <https://arxiv.org/abs/1706.03631>.
- [2] OpenProblemsInNLA, *TR-13: Equality of ranks for generic odd-order Hankel tensors*, canonical statement. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/tensor-computations/TR-13>.